

Population growth and structure

Government statisticians and demographers measure how populations have changed and make predictions about future populations and their structures. The key data they measure include the birth rate, the death rate, the infant mortality rate and net migration.

The birth rate is the number of live births per thousand of the population in one year. It is calculated by dividing the number of live births by the population and then multiplying by 1000. This means, for example, that if 800 000 babies are born one year in a country with a population of 60 million, its birth rate will be 800 000 divided by 60 million and then multiplied by 1000, that is 13. The birth rate is sometimes referred to as the crude birth rate as it provides only basic information on births. It does not, for instance, give details on the age of those who are giving birth.

The death rate is the number of deaths per thousand of the population in one year. The death rate includes the deaths of children aged under one, although countries also measure these separately in the infant mortality rate. If the birth rate exceeds the death rate, a country is said to experience a natural increase in population.

A country's population size can also be influenced by external migration that is migration into and out of the country. If immigration into the country is greater than emigration out of the country, the country is said to experience net immigration or positive net migration. For example, net migration rate of 3 would mean that a country with a population of 45m would have 135 000 ($45\text{m}/1000 \times 3$) more people coming to live in the country than leaving it.

Causes of changes in population

As countries develop, the growth in their populations tend to decline. While their death rate decreases, their birth rates fall at a greater rate. Better healthcare, education, sanitation, housing and nutrition contribute to people living longer. A decline in infant mortality, increased cost of bringing up children and an increase in women's participation in the labour force usually result in a decline in the birth rate.

Some countries with a high income per head and high development experience a natural rate of decrease in population. A number of these still experience a rise in population because they attract net immigration. People may want to move to the country to gain higher income and to enjoy a higher living standard.

Low development is often associated with high rates of population growth. This is due largely to natural increases in population size with the birth rate exceeding the death rate. There are a number of reasons why the birth rate tends to be high in countries with low development. These include the need to have children to support parents in their old age, lack of availability of methods of birth control, the relatively low costs of raising children and lack of education of women. High infant mortality rates also encourage families to have more children as they will not expect all of them to survive.

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The higher birth rate of countries with low development results in a relatively low average age of population. This creates a high proportion of dependent, non-productive members of the population. They are said to have very high dependency ratios. This means that a proportionally small working population has to produce enough goods and services to sustain not only themselves but also a large number of young people who are economically dependent upon them.

As income per head, education and healthcare start to rise, the death rate initially tends to fall more rapidly than the birth rate. Then the death rate starts to fall more slowly with the decline in the birth rate overtaking the decline in the death rate. However, high development does not eliminate population problems - it just brings new ones. The most common problem is an ageing population. This arises from a decreasing birth rate and a decreasing death rate. Again, dependency ratios are high, this time because there is a high proportion of old people who are reliant upon the productive proportion of the population for support. With people living longer, the cost of health care and pensions have been rising. To reduce the cost of pensions, a number of governments of developed countries have increased the retirement age.

The optimum population

In assessing the effects of differences in population size, economists make use of the concept of optimum population. The optimum population is said to exist when output per head is the greatest, given existing quantities of the other factors of production and the current state of technical knowledge. This is shown in Figure 51.6.

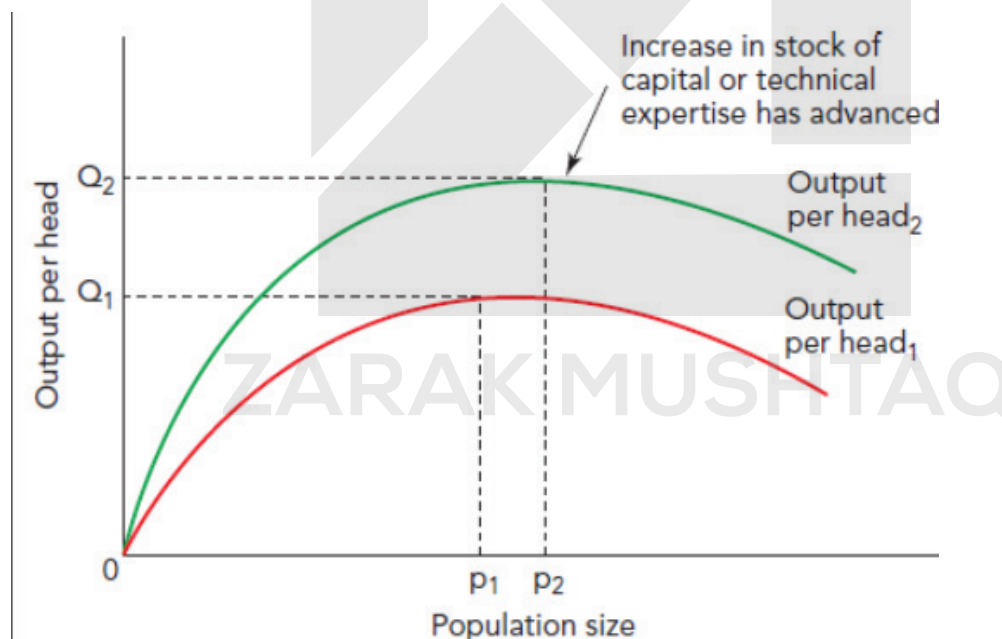


Figure 51.6: The optimum population

As the population grows, it can make better use of the stock of the other factors of production such as land and capital. This is because increasing returns are enjoyed as the population grows. If population is below the optimum level, the country can be described as underpopulated. When the

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population is beyond the optimum level and decreasing returns are being experienced, the country is said to be overpopulated. In the real world, the situation is more dynamic and the state of technical knowledge is constantly improving. The quantity of the other factors also continuously changes so that the optimum population for a country is not fixed. In addition, the criteria for assessing under population or overpopulation are purely economic and may be disputed by conservationists.

Level of urbanization

Countries with a low income per head tend to have a relatively high proportion of their populations living in rural areas. They also have rapid rates of rural-urban migration. Rural-urban migration can put pressure on the infrastructure, housing and schools in urban areas.

Most countries with a high income per head already have the majority of their populations living in urban areas. As a result, there is relatively little growth in the urban population of high-income economies. In some cases, people are moving out of urban areas to rural areas as advances in technology are allowing more people to work from home. For example, in 2019, 84% of the population of the USA was living in urban areas and the rate of rural-urban migration between 2003 and

2013 was -0.2% per year. In contrast, 30% of the population of Burkina Faso lived in urban areas in the same year and the rate of rural-urban migration was 8.1%.

Income distribution

It is a characteristic of some countries that income is unevenly distributed. This is partly because income-generating assets, especially land, are owned by a few people. As a result, there are great extremes of wealth and poverty. Income is particularly unevenly distributed in Latin America and in South Africa and Lesotho. Income is relatively evenly distributed in Scandinavian European economies.

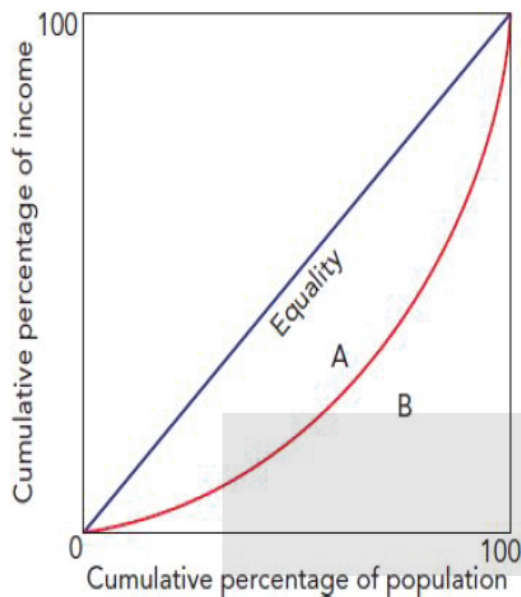
The transition of Eastern European countries, including Poland and Russia, from centrally planned to market economies has increased inequality as a small number of people have benefited enormously from the new opportunities that have resulted. Two exceptions are Slovenia and the Ukraine, which have two of the least uneven distributions of national income in Europe.

The Gini coefficient is one of the main measures of income inequality. The Lorenz curve can be used to illustrate income inequality.

Gini coefficient

The Gini coefficient is a numerical measure of the extent of inequality. Figure 51.7 shows a 45° line, the line of equality, which shows income being evenly distributed with e.g. 10% of the population receiving 10% of the income, 20% of the population would receiving 20% of the income etc. The Gini coefficient is calculated by dividing the area A by the combined areas A and B.

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The area under the line is 50% of the area of shown. 0 would mean that the income distribution is equal. (0% / 50%). There would be no gap between the actual income distribution and the line of equality. In contrast, 1 (50% / 50%), would mean there is complete inequality. All income would go to one person. Both extremes do not occur in practice. The coefficient will be between the two. The higher the figure the more unequal the distribution of income.

Table 51.1 shows the Gini coefficient for a selected number of countries in 2018.

Figure 51.7: A Lorenz curve showing the areas used to calculate the Gini coefficient

County	Gini coefficient	Country	Gini coefficient
Argentina	0.41	Russia	0.38
Belarus	0.25	Thailand	0.36
Brazil	0.54	Turkey	0.41
Costa Rica	0.48	Ukraine	0.26

Table 51.1: Gini coefficient for a selected number of countries, 2018

The Gini coefficient is sometimes expressed as the Gini Index. This is the Gini coefficient multiplied by 100, so that, for example, Argentina would have had an index of 41 in 2018.

The Lorenz curve

Figure 51.7 is a Lorenz curve. On the vertical axis, the cumulative percentage of income or wealth is plotted and on the horizontal axis the cumulative percentage of population is indicated. Table 51.2 gives an example of how the cumulative percentage of income is calculated. The 45° diagonal line shows an equal distribution of income or wealth. The actual distribution is plotted on the diagram. The further the curve is from the 45° line the more unequal the distribution.

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Percentage of the population	% share of income	Cumulative % of income
Poorest 20%	5	5
Next poorest 20%	8	13
Middle 20%	15	28
Next richest 20%	25	53
Richest 20%	47	100

Table 51.2: The distribution of income in a country

Economic structure

Employment in different sectors of the economy

As well as population growth and structure and income distribution changing as countries develop, its economic structure can also change. Countries with high income per head and high development tend to have most of their labour force employed in the tertiary sector. In contrast, economies with lower income per head and development typically have a high dependence on the primary sector. A high dependency on agricultural output and employment makes these economies vulnerable to the forces of nature. In economies that are dependent upon agricultural output for subsistence, a drought can quickly lead to famine. In those countries that are dependent upon agricultural products for their exports, a drought can wipe out their foreign currency earnings.

As an economy develops, the contribution to employment and GDP of the primary sector tends to decline. The secondary sector becomes the major source of employment and as the economy develops further, the tertiary sector usually makes the largest contribution to employment. In some cases, employment moves to a greater extent from the primary sector to the tertiary sector than to the secondary sector. This can occur when there is a growth in, for instance, tourism. It is also the case that a smaller proportion of the economy's labour force may be employed in the primary sector in a country with high development, but the total value of the sector's output may increase given higher labour and capital productivity. The different sectors of production are shown in Table 51.4.

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Sector	Description of sector
Primary sector	This covers agriculture and the extractive industries such as oil extraction, gold mining and coal mining.
Secondary sector	This includes all manufacturing industries and the construction industry.
tertiary sector	This covers all service industries such as banking, education, healthcare and tourism.

Table 51.4: The different sectors of production

Table 51.5 compares the employment composition of three countries with very high human development, according to the UN, and three countries with low human development.

Very high human development	% of labour force employed in agriculture	% of labour force employed in manufacturing	% of labour force employed in services
Australia	4	21	75
Denmark	2	18	80
UK	1	15	84
Low human development			
Democratic Republic of the Congo	35	21	44
Haiti	38	12	50
Niger	80	3	17

Table 51.5: Employment composition of selected countries

Pattern of trade at different levels of employment

In the case of a number of economies with low development and low income per head, primary products account for most of the export revenue earned. This makes them vulnerable in their trading relations because demand and especially supply can change significantly. For example, demand may decrease due to a health scare, and supply may increase due to a good harvest. These

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changes mean that the market for primary products is subject to frequent and large fluctuations in price.

In addition, the demand for many primary products, especially food, is income inelastic. This means that as world income rises there is little impact upon the demand for primary products. In contrast, demand for manufactured goods is income elastic. This means that primary goods tend to become relatively cheaper than manufactured goods. As a result, over time there is a tendency for the terms of trade of primary goods to decline relative to manufactured goods. Those countries that are dependent upon agricultural products may receive relatively low prices for their exports while having to pay relatively high prices for their imports.

Countries with high development and high income tend to export mainly manufactured goods and services. They also tend to export a wide range of products. Some low-income countries, in contrast, rely heavily on exporting a narrow range of products. For example, in 2019 exports of coffee accounted for 35% of Ethiopia's export revenue.



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